

Lab 12: Hypothesis Testing and ANOVA Capstone

Level: Advanced Mapping: LO6 | K5 | A1-A10 Concepts: H0/H1, p-value, t-test, ANOVA, diagnostics, remediation

Scenario

A health-policy team compares groups and proposes a defensible model update.

Goal

A t-test, ANOVA, post-hoc comparison and revised analysis decision.

Files in this folder

- lab-12.R - completed guided R script
- verify.R - repeatable acceptance check
- lab-workbook.xlsx - brief, input data, controls and evidence record
- input-data.csv - portable fallback if readxl is unavailable

Detailed step-by-step

1. Write H0, H1 and alpha before running any test.
2. Run the specified two-group t-test and report estimates, interval and p-value.
3. Run one-way ANOVA across three or more groups.
4. If warranted, run TukeyHSD and identify supported pairwise differences.
5. Diagnose one unintended outcome and implement a justified script update.

Run and verify

1. Set the RStudio working directory to this lab folder.
2. Open lab-12.R and read the header, inputs and validation rules.
3. Run the complete script with `source("lab-12.R")`.
4. Confirm the Console ends with LAB VERIFIED: outputs created and checks passed.
5. Run `source("verify.R")` and retain the verification message.
6. Open analysis_output.csv and compare the evidence with the workbook Control sheet.
7. Complete the workbook Evidence sheet and save it without changing the filename.

Acceptance criteria

- [] Every conclusion uses reject/fail-to-reject language correctly.
- [] ANOVA conclusion does not overstate which groups differ.
- [] The updated script reruns successfully and the audit note explains the change.

Troubleshooting

- there is no package called 'readxl': the script automatically uses input-data.csv; install readxl later if Excel import is required.
- cannot open file: confirm RStudio's working directory is this exact lab folder.
- object not found: run the script from the top so imports and validations occur before analysis.
- Verification fails: inspect the first error, compare column names with Input_Data, and rerun from a clean session.

Evidence to submit

Completed Excel Evidence sheet, R script, generated output files and the final Console verification message.